

Лабораторная работа №08

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Цель работы

Приобретение практических навыков администрирования сетевых подсистем

На виртуальной машине **server** войдём под нашим пользователем и откроем терминал. Перейдём в режим суперпользователя и установим необходимые для работы пакеты (рис. @fig-1, @fig-2):

```
[frank@server ~]$ sudo -i

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.

[sudo] password for frank:
[root@server ~]# dnf -y install postfix
Extra Packages for Enterprise Linux 9 - x86_64 6.8 kB/s | 37 kB 00:05
Extra Packages for Enterprise Linux 9 - x86_64 1.1 MB/s | 20 MB 00:18
Extra Packages for Enterprise Linux 9 openh264 167 B/s | 993 B 00:05
Rocky Linux 9 - BaseOS 285 B/s | 4.1 kB 00:14
Rocky Linux 9 - BaseOS 151 kB/s | 2.5 MB 00:17
Rocky Linux 9 - AppStream 413 B/s | 4.5 kB 00:11
Rocky Linux 9 - AppStream 497 kB/s | 9.5 MB 00:19
Rocky Linux 9 - Extras 260 B/s | 2.9 kB 00:11
Rocky Linux 9 - Extras 1.1 kB/s | 17 kB 00:15
Dependencies resolved.
=====
Package Architecture Version Repository Size
=====
Installing:
postfix x86_64 2:3.5.25-1.el9 appstream 1.5 M
Transaction Summary
=====
Install 1 Package

Total download size: 1.5 M
Installed size: 4.4 M
```

Рис. 1: Этап 1

Установка Postfix на сервере

```
complete:
[root@server ~]# dnf -y install s-nail
Last metadata expiration check: 0:05:25 ago on Wed 22 Oct 2025 09:32:43 PM UTC.
Dependencies resolved.
=====
Package                        Architecture      Version           Size              Repository
=====
Installing:
s-nail                         x86_64            14.9.22-6.el9    621 k             appstream

Transaction Summary
=====
Install 1 Package

Total download size: 621 k
Installed size: 1.1 M
Downloading Packages:
s-nail-14.9.22-6.el9.x86_64.rpm                                109 kB/s | 621 kB    00:05
-----
Total                                                         53 kB/s | 621 kB    00:11
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing                :                                     1/1
  Running scriptlet: s-nail-14.9.22-6.el9.x86_64                1/1
  Installing             : s-nail-14.9.22-6.el9.x86_64          1/1
  Running scriptlet: s-nail-14.9.22-6.el9.x86_64                1/1
  Verifying               : s-nail-14.9.22-6.el9.x86_64          1/1
Installed:
```

Рис. 2: Этап 2

Сконфигурируем межсетевой экран, разрешив работать службе протокола SMTP. Восстановим контекст безопасности в SELinux и запустим Postfix (рис. @fig-3):

```
[root@server ~]# firewall-cmd --add-service=smtp
success
[root@server ~]# firewall-cmd --add-service=smtp --permanent
success
[root@server ~]# firewall-cmd --list-services
cockpit dhcpv6-client smtp ssh
[root@server ~]# restorecon -vR /etc
Relabeled /etc/sysconfig/network-scripts/ifcfg-eth1 from unconfined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:net_conf_t:s0
[root@server ~]# systemctl enable postfix
Created symlink /etc/systemd/system/multi-user.target.wants/postfix.service → /usr/lib/systemd/system/postfix.service.
[root@server ~]# systemctl start postfix
[root@server ~]#
```

Рис. 3: Этап 3

Посмотрим и изменим параметры Postfix, заменив значение параметра myorigin на значение параметра mydomain (рис. @fig-4):

Проверим корректность конфигурационного файла и перезагрузим Postfix (рис. @fig-5):

```
[root@server ~]# postconf myorigin
myorigin = $myhostname
[root@server ~]# postconf mydomain
mydomain = frank.net
[root@server ~]# postconf -e 'myorigin = $mydomain'
postconf: fatal: missing '=' after attribute name: "???myorigin"
[root@server ~]# postconf myorigin
myorigin = $myhostname
[root@server ~]#
```

Рис. 4: Этап 4

```
[root@server ~]# postconf -n
alias_database = hash:/etc/aliases
alias_maps = hash:/etc/aliases
command_directory = /usr/sbin
compatibility_level = 2
daemon_directory = /usr/libexec/postfix
data_directory = /var/lib/postfix
debug_peer_level = 2
debugger_command = PATH=/bin:/usr/bin:/usr/local/bin:/usr/X11R6/bin ddd $daemon_directory/$process_name
                    $process_id & sleep 5
html_directory = no
inet_interfaces = localhost
inet_protocols = all
mail_owner = postfix
mailq_path = /usr/bin/mailq.postfix
manpage_directory = /usr/share/man
meta_directory = /etc/postfix
mydestination = $myhostname, localhost.$mydomain, localhost
mydomain = frank.net
newaliases_path = /usr/bin/newaliases.postfix
queue_directory = /var/spool/postfix
readme_directory = /usr/share/doc/postfix/README_FILES
sample_directory = /usr/share/doc/postfix/samples
sendmail_path = /usr/sbin/sendmail.postfix
setgid_group = postdrop
shlib_directory = /usr/lib64/postfix
smtp_tls_CAfile = /etc/pki/tls/certs/ca-bundle.crt
```

Рис. 5: Этап 5

```
[root@server ~]# postconf -e 'mydomain = frank.net'
[root@server ~]# postconf inet_protocols
inet_protocols = all
[root@server ~]# postconf -e 'inet_protocols = ipv4'
[root@server ~]# postfix check
[root@server ~]# systemctl reload postfix
[root@server ~]#
```

Рис. 6: Этап 6

На сервере под учётной записью пользователя отправим себе письмо, используя утилиту mail (рис. @fig-6):

Запустим мониторинг работы почтовой службы и посмотрим, что произошло с нашим сообщением (рис. @fig-7):

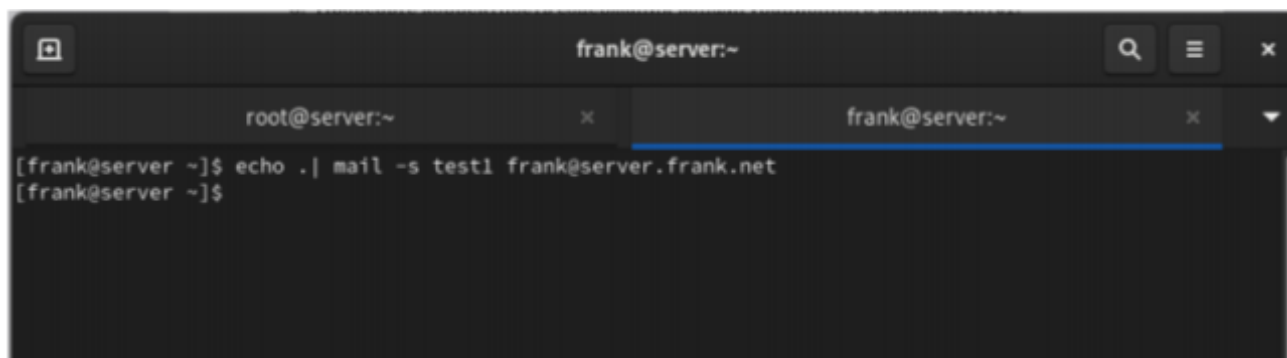


Рис. 7: Этап 7

На виртуальной машине client установим необходимые для работы пакеты (рис. @fig-8, @fig-9):

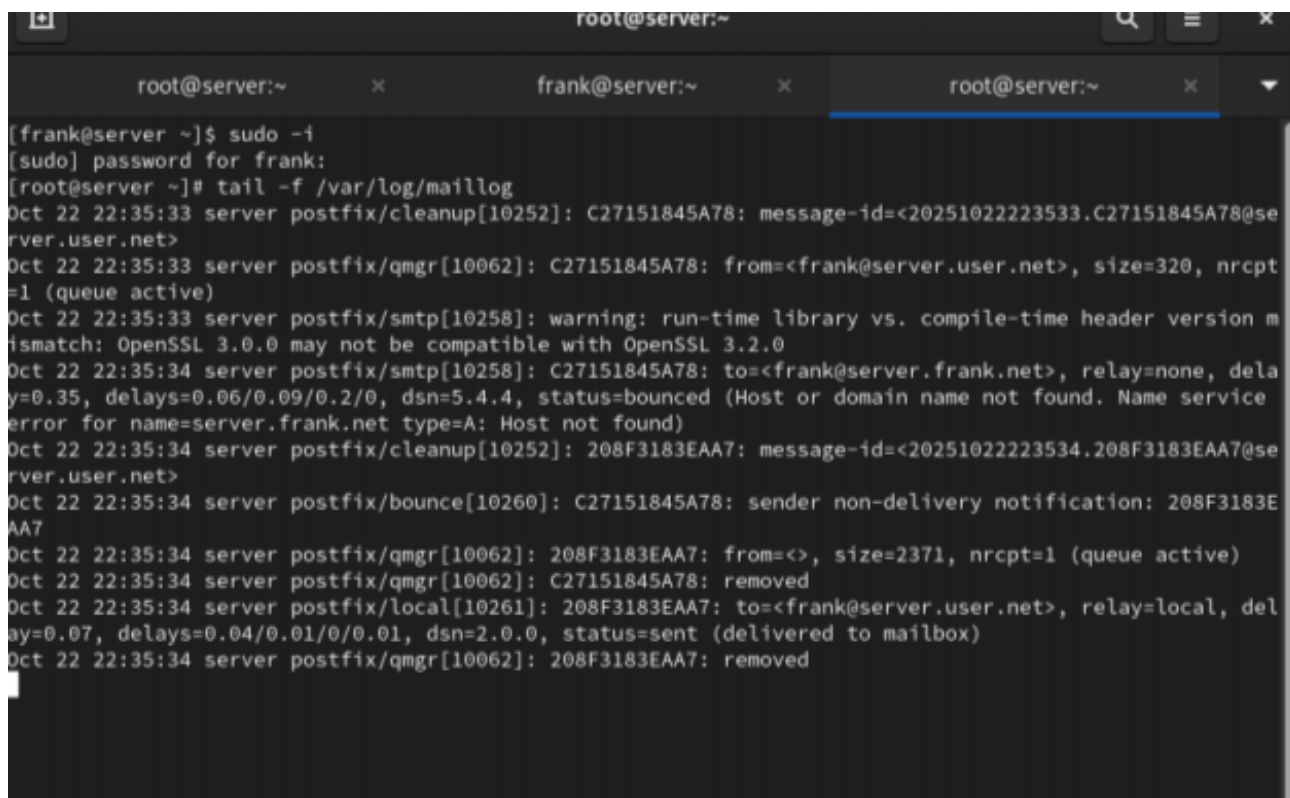


Рис. 8: Этап 8

Установка Postfix на клиенте

Настроим Postfix на клиенте и отправим тестовое письмо (рис. @fig-10):

```

#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.

[sudo] password for frank:
[root@client ~]# dnf -y install postfix
Last metadata expiration check: 0:04:58 ago on Wed 22 Oct 2025 11:11:26 PM UTC.
Dependencies resolved.
=====
Package           Architecture Version           Repository        Size
=====
Installing:
postfix           x86_64          2:3.5.25-1.el9    appstream         1.5 M

Transaction Summary
=====
Install 1 Package

Total download size: 1.5 M
Installed size: 4.4 M
Downloading Packages:
postfix-3.5.25-1.el9.x86_64.rpm        6.2 MB/s | 1.5 MB    00:00
=====

```

Рис. 9: Этап 9

```

#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.

[sudo] password for frank:
[root@client ~]# dnf -y install postfix
Last metadata expiration check: 0:04:58 ago on Wed 22 Oct 2025 11:11:26 PM UTC.
Dependencies resolved.
=====
Package           Architecture Version           Repository        Size
=====
Installing:
postfix           x86_64          2:3.5.25-1.el9    appstream         1.5 M

Transaction Summary
=====
Install 1 Package

Total download size: 1.5 M
Installed size: 4.4 M
Downloading Packages:
postfix-3.5.25-1.el9.x86_64.rpm        6.2 MB/s | 1.5 MB    00:00
=====

```

Рис. 10: Этап 10

На сервере настроим параметры сетевых интерфейсов и перезапустим Postfix (рис. @fig-11):

```
[root@client ~]# dnf install s-nail
Last metadata expiration check: 1:39:11 ago on Sat 25 Oct 2025 12:07:43 PM UTC.
Package s-nail-14.9.22-6.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@client ~]#
```

Рис. 11: Этап 11

С клиента отправим письмо на свой доменный адрес и посмотрим очередь сообщений (рис. @fig-12):

```
[root@client ~]# postconf inet_protocols
inet_protocols = all
[root@client ~]# postconf -e 'inet_protocols = ipv4'
[root@client ~]# systemctl enable postfix
Created symlink /etc/systemd/system/multi-user.target.wants/postfix.service → /usr/lib/systemd/system/postfix.service.
[root@client ~]# systemctl start postfix
[root@client ~]#
```

Рис. 12: Этап 12

Пропишем MX-запись в файлах прямой и обратной DNS-зон (рис. @fig-13):

Восстановим контекст безопасности, перезапустим DNS и отправим сообщения из очереди (рис. @fig-14):

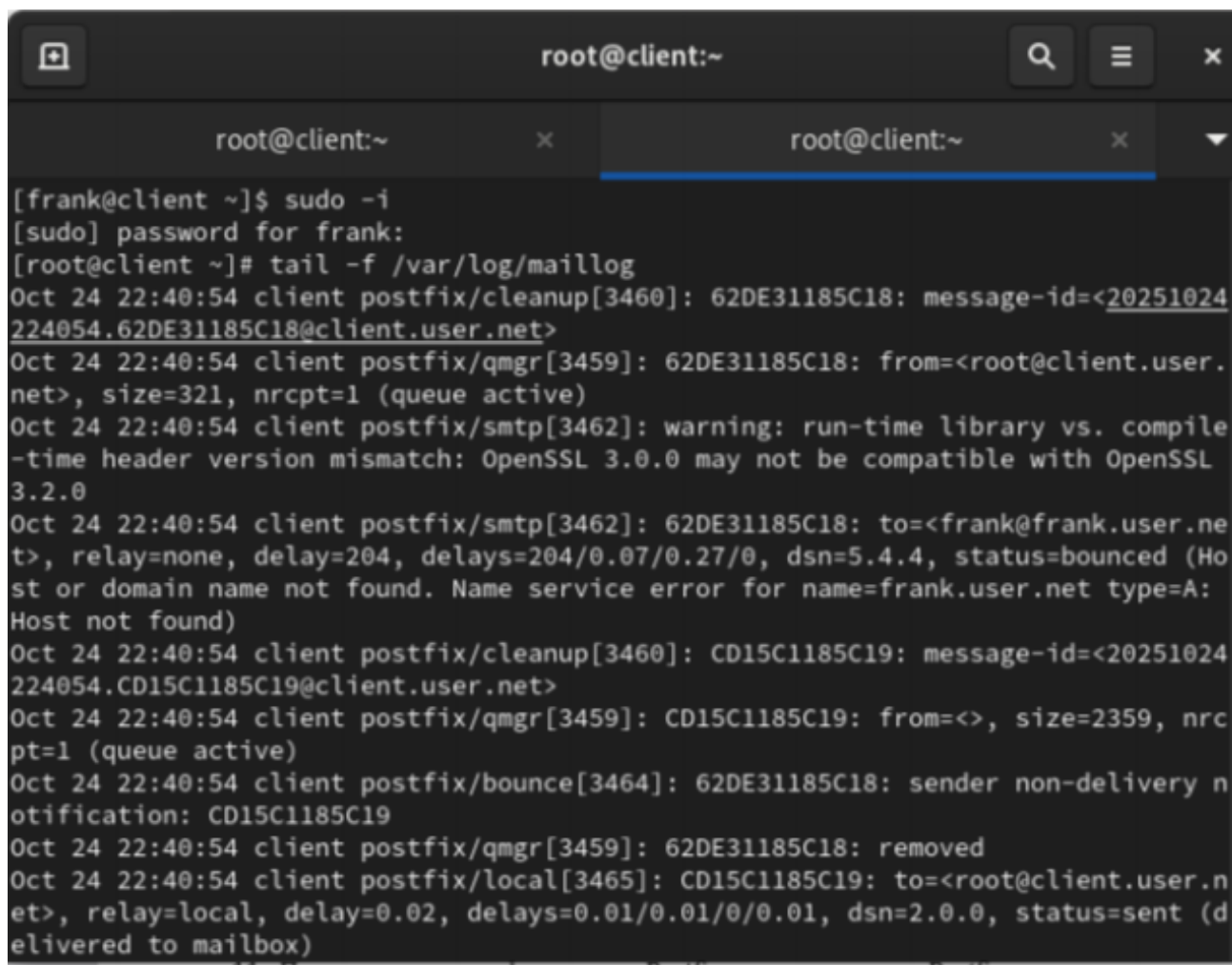
Настроим скрипты для автоматического развёртывания и добавим записи в Vagrantfile (рис. @fig-15):

Этап выполнения работы №16

Этап выполнения работы №17

Этап выполнения работы №18

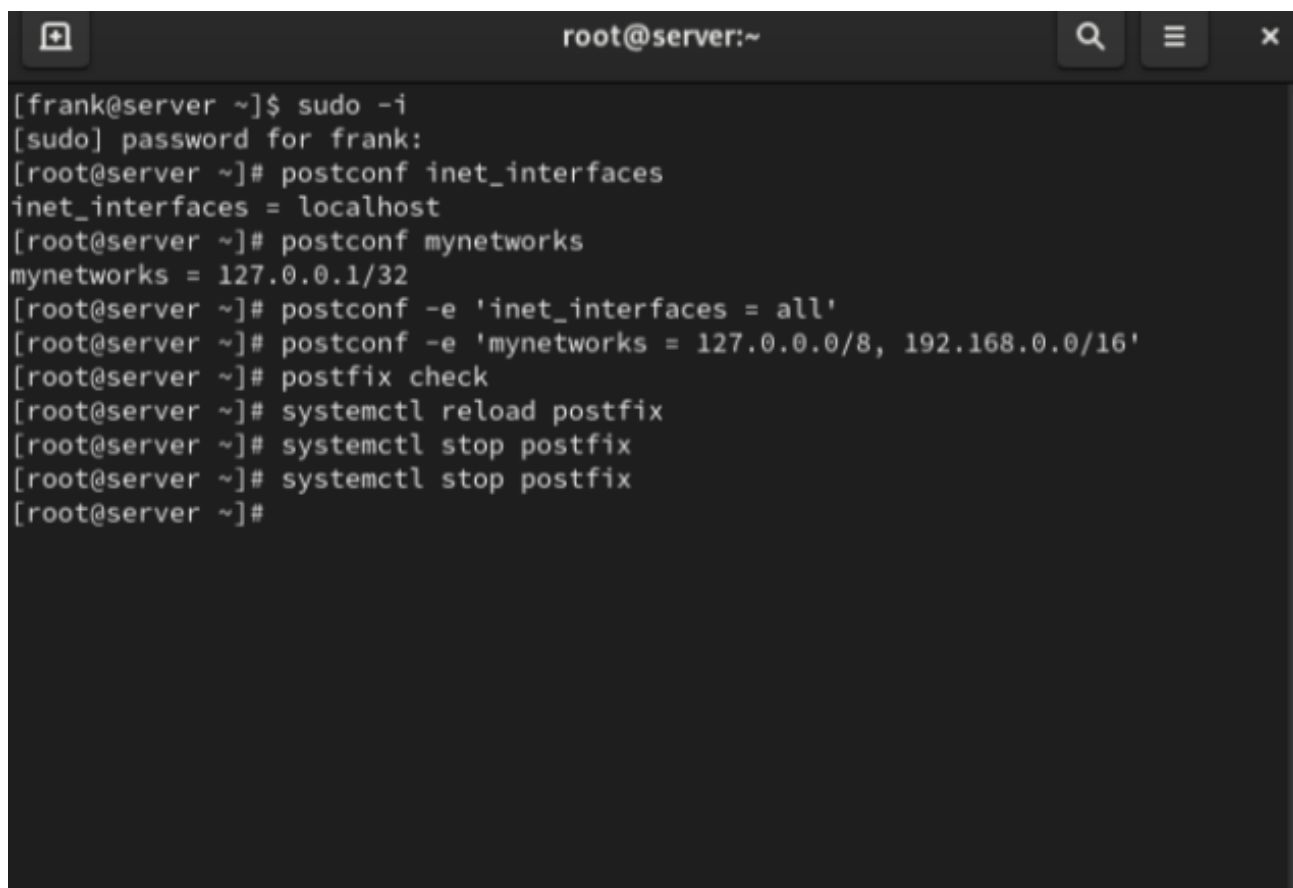
Этап выполнения работы №19



A terminal window titled 'root@client:~' showing the output of a 'tail -f /var/log/maillog' command. The output displays several mail-related events, including message cleanup, queue management, and a bounced email notification. The window has a dark theme and standard terminal icons at the top.

```
[frank@client ~]$ sudo -i
[sudo] password for frank:
[root@client ~]# tail -f /var/log/maillog
Oct 24 22:40:54 client postfix/cleanup[3460]: 62DE31185C18: message-id=<20251024224054.62DE31185C18@client.user.net>
Oct 24 22:40:54 client postfix/qmgr[3459]: 62DE31185C18: from=<root@client.user.net>, size=321, nrcpt=1 (queue active)
Oct 24 22:40:54 client postfix/smtp[3462]: warning: run-time library vs. compile-time header version mismatch: OpenSSL 3.0.0 may not be compatible with OpenSSL 3.2.0
Oct 24 22:40:54 client postfix/smtp[3462]: 62DE31185C18: to=<frank@frank.user.net>, relay=none, delay=204, delays=204/0.07/0.27/0, dsn=5.4.4, status=bounced (Host or domain name not found. Name service error for name=frank.user.net type=A: Host not found)
Oct 24 22:40:54 client postfix/cleanup[3460]: CD15C1185C19: message-id=<20251024224054.CD15C1185C19@client.user.net>
Oct 24 22:40:54 client postfix/qmgr[3459]: CD15C1185C19: from=<>, size=2359, nrcpt=1 (queue active)
Oct 24 22:40:54 client postfix/bounce[3464]: 62DE31185C18: sender non-delivery notification: CD15C1185C19
Oct 24 22:40:54 client postfix/qmgr[3459]: 62DE31185C18: removed
Oct 24 22:40:54 client postfix/local[3465]: CD15C1185C19: to=<root@client.user.net>, relay=local, delay=0.02, delays=0.01/0.01/0/0.01, dsn=2.0.0, status=sent (delivered to mailbox)
```

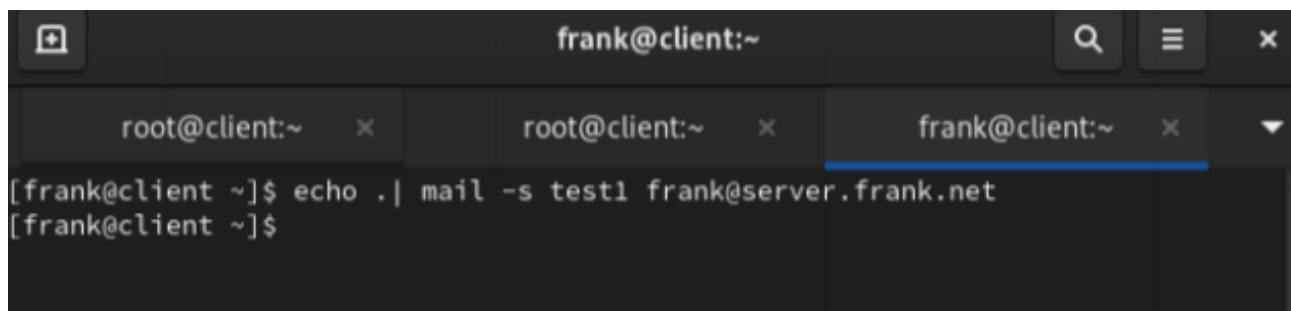
Рис. 13: Этап 13



A terminal window titled 'root@server:~' showing the execution of Postfix configuration commands. The user 'frank' uses 'sudo' to become 'root'. The commands modify 'inet_interfaces' to 'all' and 'mynetworks' to include '192.168.0.0/16', then check the configuration and reload the Postfix service. The window has a dark theme and standard terminal icons at the top.

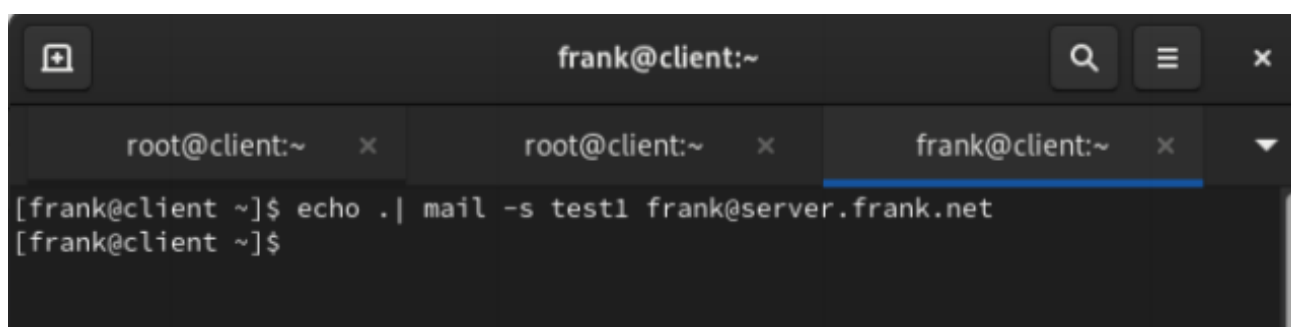
```
[frank@server ~]$ sudo -i
[sudo] password for frank:
[root@server ~]# postconf inet_interfaces
inet_interfaces = localhost
[root@server ~]# postconf mynetworks
mynetworks = 127.0.0.1/32
[root@server ~]# postconf -e 'inet_interfaces = all'
[root@server ~]# postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'
[root@server ~]# postfix check
[root@server ~]# systemctl reload postfix
[root@server ~]# systemctl stop postfix
[root@server ~]# systemctl stop postfix
[root@server ~]#
```

Рис. 14: Этап 14

A terminal window titled 'frank@client:~' with search, menu, and close icons. It has three tabs: 'root@client:~', 'root@client:~', and 'frank@client:~'. The active tab shows the command '[frank@client ~]\$ echo .| mail -s test1 frank@server.frank.net' and the prompt '[frank@client ~]\$' on the next line.

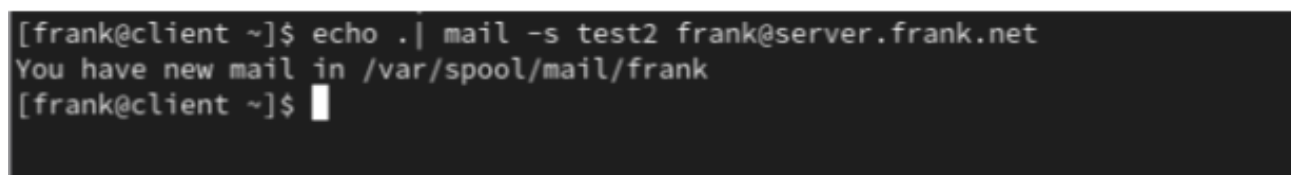
```
frank@client:~
[frank@client ~]$ echo .| mail -s test1 frank@server.frank.net
[frank@client ~]$
```

Рис. 15: Этап 15

A terminal window titled 'frank@client:~' with search, menu, and close icons. It has three tabs: 'root@client:~', 'root@client:~', and 'frank@client:~'. The active tab shows the command '[frank@client ~]\$ echo .| mail -s test1 frank@server.frank.net' and the prompt '[frank@client ~]\$' on the next line.

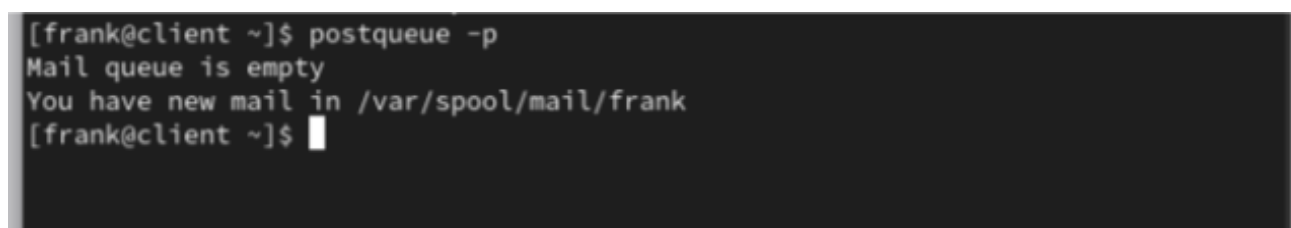
```
frank@client:~
[frank@client ~]$ echo .| mail -s test1 frank@server.frank.net
[frank@client ~]$
```

Рис. 16: Этап 16

A terminal window showing the command '[frank@client ~]\$ echo .| mail -s test2 frank@server.frank.net'. The output is 'You have new mail in /var/spool/mail/frank'. The prompt '[frank@client ~]\$' is followed by a cursor.

```
[frank@client ~]$ echo .| mail -s test2 frank@server.frank.net
You have new mail in /var/spool/mail/frank
[frank@client ~]$
```

Рис. 17: Этап 17

A terminal window showing the command '[frank@client ~]\$ postqueue -p'. The output is 'Mail queue is empty' and 'You have new mail in /var/spool/mail/frank'. The prompt '[frank@client ~]\$' is followed by a cursor.

```
[frank@client ~]$ postqueue -p
Mail queue is empty
You have new mail in /var/spool/mail/frank
[frank@client ~]$
```

Рис. 18: Этап 18


```
root@server:~/named/master
frank@server:~ x frank@server:~ x root@server:~/na... x
IN SOA @ server.frank.net. (
    2020110500 ; serial
    1D ; refresh
    1H ; retry
    1W ; expire
    3H ) ; minimum
NS
A 192.168.1.1
MX 10 mail.frank.net.
$ORIGIN frank.net.
server A 192.168.1.1
ns A 192.168.1.1
dhcp A 192.168.1.1
www A 192.168.1.1
mail A 192.168.1.1
-- INSERT -- 17,16 Bot
```

Рис. 19: Этап 19

Этап выполнения работы №20

Этап выполнения работы №21

Этап выполнения работы №22

Этап выполнения работы №23

Этап выполнения работы №24

Этап выполнения работы №25

Этап выполнения работы №26

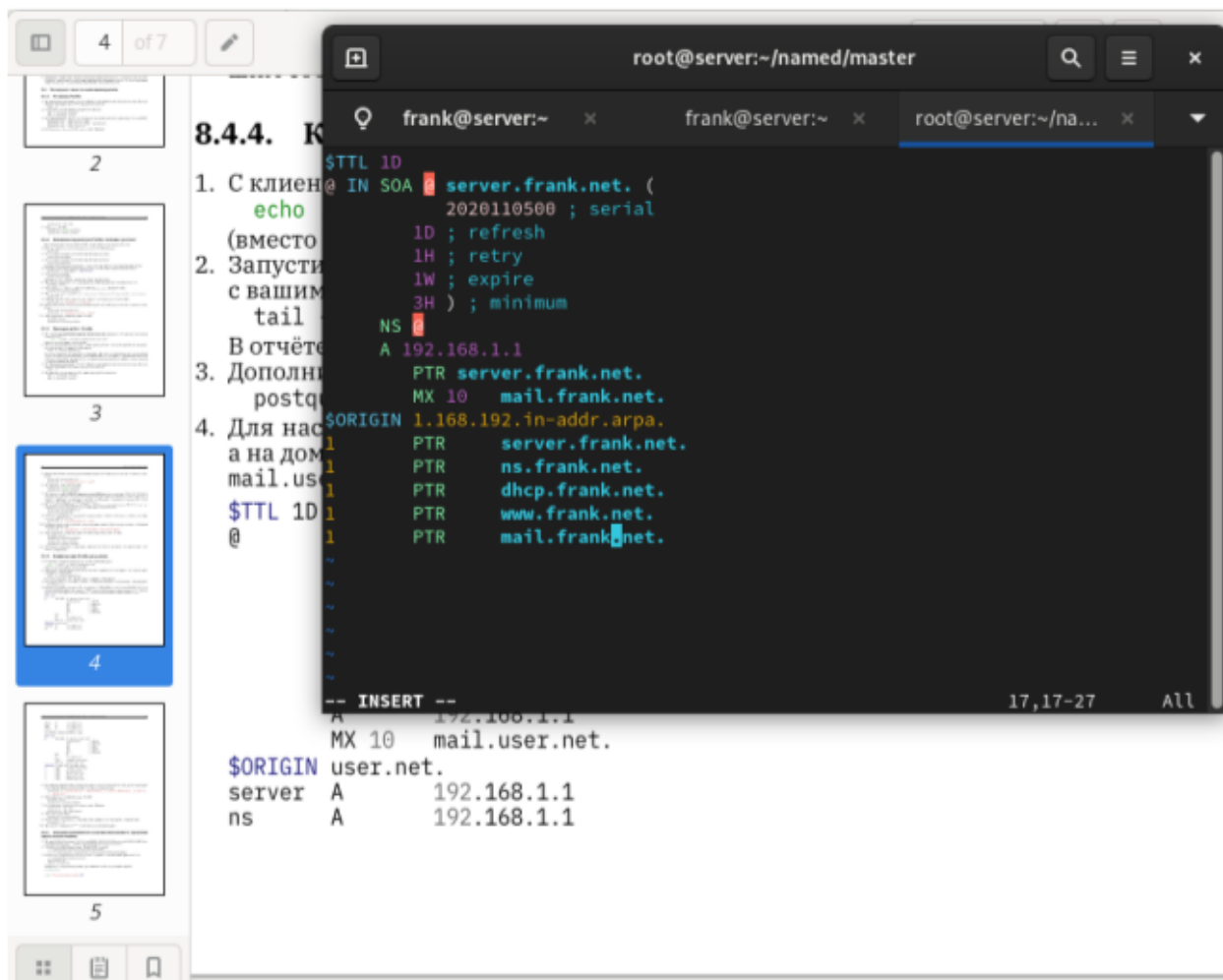


Рис. 20: Этап 20

```
[root@server ~]# postconf -e 'mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain'
[root@server ~]#
```

Рис. 21: Этап 21

```
[root@server /]# restorecon -vR /etc
[root@server /]# postfix check
[root@server /]# systemctl reload postfix
[root@server /]# restorecon -vR /etc
[root@server /]# restorecon -vR /var/named
restorecon: lstat(/var/named) failed: No such file or directory
[root@server /]# cd var
[root@server var]# ls
account  cache  db      ftp      kerberos  local  log      nis      preserve  spool  yp
```

Рис. 22: Этап 22

```
[root@server ~]# systemctl restart named
Failed to restart named.service: Unit named.service not found.
[root@server ~]# postqueue -f
bash: postqueue: command not found...
[root@server ~]# postqueue -f
[root@server ~]#
```

Рис. 23: Этап 23

```
[root@client ~]# echo . | mail -s test2 frank@frank.net
[root@client ~]#
```

Рис. 24: Этап 24

```
[root@client ~]# tail -F /var/log/maillog
Oct 24 23:40:20 client postfix/qmgr[3459]: B33FC1185C19: from=<>, size=2371, nrc
pt=1 (queue active)
Oct 24 23:40:20 client postfix/bounce[3855]: 79F9211C1723: sender non-delivery n
otification: B33FC1185C19
Oct 24 23:40:20 client postfix/qmgr[3459]: 79F9211C1723: removed
Oct 24 23:40:20 client postfix/local[3856]: B33FC1185C19: to=<frank@client.user.
net>, relay=local, delay=0.01, delays=0/0.01/0/0, dsn=2.0.0, status=sent (delive
red to mailbox)
Oct 24 23:40:20 client postfix/qmgr[3459]: B33FC1185C19: removed
Oct 25 10:35:01 client postfix/pickup[4294]: 9CEAC11C1723: uid=0 from=<root>
Oct 25 10:35:01 client postfix/cleanup[4495]: 9CEAC11C1723: message-id=<20251025
103501.9CEAC11C1723@client.user.net>
Oct 25 10:35:01 client postfix/qmgr[3459]: 9CEAC11C1723: from=<root@client.user.
net>, size=316, nrcpt=1 (queue active)
Oct 25 10:35:01 client postfix/smtp[4497]: warning: run-time library vs. compile
-time header version mismatch: OpenSSL 3.0.0 may not be compatible with OpenSSL
3.2.0
Oct 25 10:35:37 client postfix/smtp[4497]: connect to za-smtp-inbound-2.mimecast
```

Рис. 25: Этап 25

```
[root@server server]# ls
01-dummy.sh 02-forward.sh
[root@server server]# touch mail.sh
[root@server server]# chmod +x mail.sh
[root@server server]#
```

Рис. 26: Этап 26

```
root@server:/vagrant/provision/server
frank@server:~ x frank@server:~ x root@server:/vag... x
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install postfix
dnf -y install s-nail
echo "Copy configuration files"
#cp -R /vagrant/provision/server/mail/etc/* /etc
echo "Configure firewall"
firewall-cmd --add-service=smtp --permanent
firewall-cmd --reload
restorecon -vR /etc
echo "Start postfix service"
systemctl enable postfix
systemctl start postfix
echo "Configure postfix"
postconf -e 'mydomain = user.net'
postconf -e 'myorigin = $mydomain'
postconf -e 'inet_protocols = ipv4'
postconf -e 'inet_interfaces = all'
postconf -e 'mydestination = $myhostname, localhost.$mydomain, localhost,
$mydomain'
postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'
-- INSERT -- 3,1 Top
```

Рис. 27: Этап 27

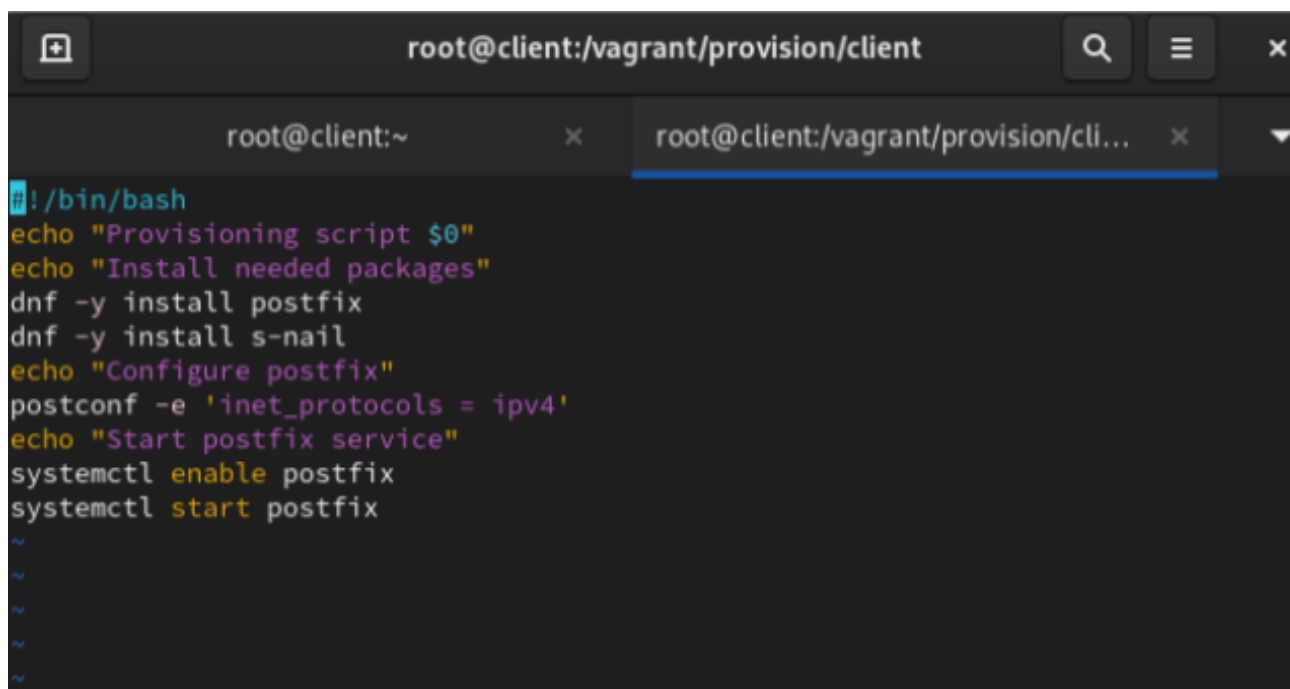
Этап выполнения работы №27

Этап выполнения работы №28

```
[root@client client]# ls
01-dummy.sh  01-routing.sh  mail.sh
[root@client client]#
```

Рис. 28: Этап 28

Этап выполнения работы №29



The screenshot shows a terminal window with the title bar 'root@client:/vagrant/provision/client'. The terminal content is as follows:

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install postfix
dnf -y install s-nail
echo "Configure postfix"
postconf -e 'inet_protocols = ipv4'
echo "Start postfix service"
systemctl enable postfix
systemctl start postfix
~
~
~
~
```

Рис. 29: Этап 29

Этап выполнения работы №30

Этап выполнения работы №31

Выводы

В ходе выполнения лабораторной работы были приобретены практические навыки

Спасибо за внимание!

Вопросы?

```

server.vm.provision "server http",
  type: "shell",
  preserve_order: true,
  path: "provision/server/http.sh"
server.vm.provision "server mysql",
  type: "shell",
  preserve_order: true,
  path: "provision/server/mysql.sh"
server.vm.provision "server firewall",
  type: "shell",
  preserve_order: true,
  path: "provision/server/firewall.sh"
server.vm.provision "server mail",
  type: "shell",
  preserve_order: true,
  path: "provision/server/mail.sh"

```

Рис. 30: Этап 30

Fichier	Modifier	Affichage
		<pre> client.vm.provision "client dummy", type: "shell", preserve_order: true, path: "provision/client/01-dummy.sh" client.vm.provision "client routing", type: "shell", preserve_order: true, run: "always", path: "provision/client/01-routing.sh" client.vm.provision "client mail", type: "shell", preserve_order: true, path: "provision/client/mail.sh" </pre>
		<pre> client.vm.provider :virtualbox do v v.linked_clone = true # Customize the amount of memory on the VM v.memory = 1024 </pre>

Рис. 31: Этап 31